AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

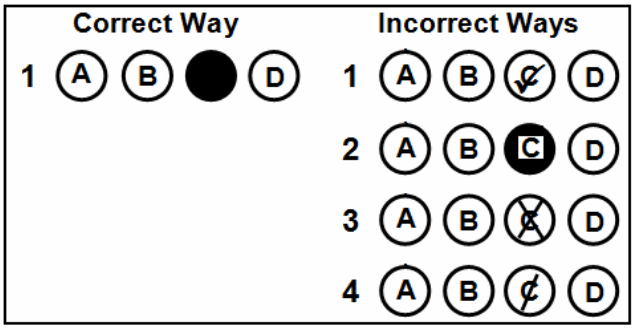
MAY 2012

General Mathematics Paper I

Time allowed: 40 minutes Marks 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. 250 % of 25 is equal to

   A. 122.5
   B. 62.5
   C. 12.25
   D. 6.25

2. If $5 : 3 :: 3 : c$, then c is equal to

   A. 5
   B. 3
   C. $\frac{5}{9}$
   D. $\frac{9}{5}$

3. If $x^2 y = k$, then

   A. $y \propto x^2$
   B. $y \propto \frac{1}{x^2}$
   C. $y \propto \sqrt{x}$
   D. $y \propto \frac{1}{\sqrt{x}}$

4. The amount of Zakat payable on a total savings of Rs. 40,000 is equal to Rs.
   [Note: Zakat rate = 2.5 % of total savings]

   A. 1,000
   B. 1,500
   C. 1,600
   D. 2,000

5. The property of Mr. Saleh is divided between his son and daughter in the ratio 2 : 1.
   The fraction of property that daughter gets is

   A. $\frac{1}{2}$
   B. $\frac{1}{3}$
   C. $\frac{2}{3}$
   D. $\frac{2}{1}$

6. A story book was sold for Rs. 60 having cost price Rs. 50. The profit percent is

   A. 20 %
   B. 16.67 %
   C. 10 %
   D. 5 %

7. If S is the selling price, C is the cost price and L is the loss, then the relation between S, C and L is

   A. $S + C = L$
   B. $S + L = C$
   C. $\frac{S + C}{2} = L$
   D. $C + L = S$

8. If marked price of a jug is Rs. 250 and discount offered is 30 % , then the selling price of the jug is Rs.

   A. 175
   B. 185
   C. 225
   D. 325

9. On a certain day, if 100 US dollars is equal to 72.45 Euro, then 20 US dollars will be equal to

   A. 14.49 Euro
   B. 13.49 Euro
   C. 12.49 Euro
   D. 11.49 Euro

10. If Rs. 30,000 is invested in a scheme having simple interest rate of 10 %, then the amount of markup in 10 years will be Rs.

    A. 3,000
    B. 6,000
    C. 30,000
    D. 60,000

11. A bank offers a car on leasing. The cost of the car is Rs. 500,000 and according to plan the car is delivered on 20 % down payment and 20 equal monthly instalments. Each monthly instalment will be Rs.

    A. 25,000
    B. 20,000
    C. 15,000
    D. 10,000

12. A promise of compensation for potential future losses in exchange for a periodic payment is called

    A. principal amount.
    B. down payment.
    C. insurance.
    D. leasing.

13. A tax based on the cost of the item purchased and collected directly from buyers is called

    A. excise tax.
    B. income tax.
    C. property tax.
    D. sales tax.

14. The price of a camera before tax is Rs. 9,000. If General Sales Tax (GST) is levied at a rate of 8 % , then the amount of sales tax will be Rs.

    A. 630
    B. 640
    C. 720
    D. 740

15. A person works 160 hours in a month at a rate of 100 rupees per hour. If he worked 20 additional hours at the rate of Rs. 200 per hour, then his total salary for the month will be Rs.

    A. 24,000
    B. 20,000
    C. 16,000
    D. 4,000

16. The given table shows a telephone bill of Ms. Azra for the month of April.

Charges Detail	Amount in Rs.
Line Rent	174
Package Charges	199
Local Call Charges	264
Mobile Call Charges	?
Value Added Tax	176
Withholding Tax	8
Total Bill	**1088**

    The mobile call charges for the month of April is Rs.

    A. 250
    B. 257
    C. 260
    D. 267

17. $\left(a^{2}\right)^{-3}$ is equal to

A. a^{-6}
B. a^{-1}
C. a^{-8}
D. $-a^{-6}$

18. The characteristic of log 0.001 is

A. 3
B. 2
C. -2
D. -3

19. $a^{0} \times a^{\frac{1}{x}}$ is equal to

A. 1
B. a
C. $a^{\frac{1}{x}}$
D. $a^{\frac{2}{x}}$

20. The next term in the sequence 1, 1, 2, 4, 3, 9, 4, 16, 5 will be

A. 18
B. 20
C. 25
D. 30

21. The geometric mean of a^{2} and a^{4} is

A. a^{3}
B. a^{4}
C. a^{6}
D. a^{8}

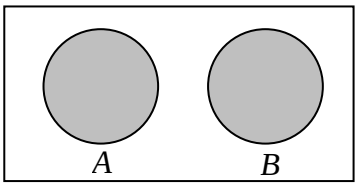
22. If both the first term and the common difference of an arithmetic progression are 7, then the nth term will be

A. $7n - 14$
B. $7n - 7$
C. $7n + 7$
D. $7n$

23. If $A = \{0\}$ and $B = \{1, 2\}$, then $A \times B$ is equal to

A. $\{0\}$
B. $\{0, 1, 2\}$
C. $\{(0, 1), (0, 2)\}$
D. $\{(0, 1), (0, 2), (1, 0), (2, 0)\}$

24. In the given Venn diagram the shaded portion represents



A. $A \cup B$
B. $A \cap B$
C. $A^C \cup B^C$
D. $A^C \cap B^C$

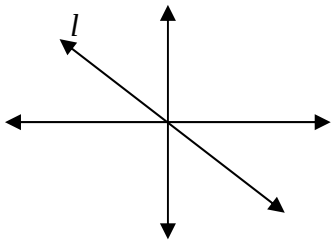
25. If $A = \{a, b\}$ and $B = \{2, 3\}$, then one relation from A to B is

A. $\{(a, b), (2, 3)\}$
B. $\{(a, 3), (2, b)\}$
C. $\{(a, 2), (a, 3)\}$
D. $\{(2, a), (3, b)\}$

26. If f is a function defined as $f = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$, then the range of f is

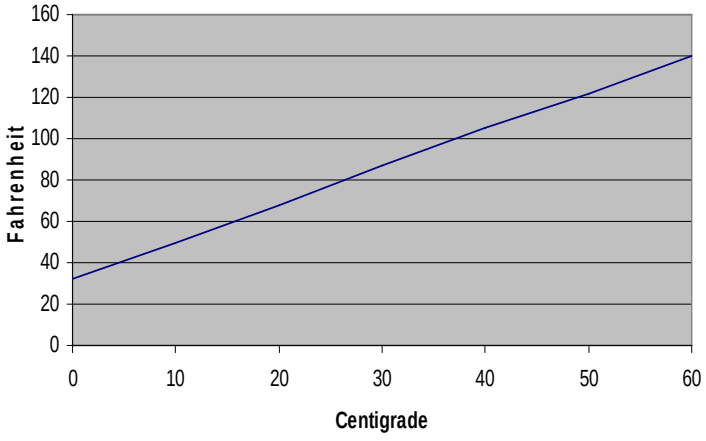
A. $\{1, 2, 3, 4\}$
B. $\{1, 1, 2, 3\}$
C. $\{1, 2, 3\}$
D. $\{2, 3\}$

27. The line l in the given graph can be represented by



A. $y = c$
B. $x = c$
C. $y = mx$
D. $y = mx + c$

28. The given graph shows the relationship between centigrade scale (C) and Fahrenheit scale (F). The temperature of $50\,°C$ is equal to



A. 100 °F
B. 110 °F
C. 112 °F
D. 122 °F

29. The following table shows the number of ice-creams sold at a shop in a certain week of May.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
110	120	110	130	80	200	220



The median of the above data is

A. 110
B. 120
C. 130
D. 220

30. The following table shows the time in minutes taken to serve people in a hotel.

Time (in minutes)	Number of People
5 – 10	15
11 – 16	20
17 – 22	25
23 – 28	15

The number of people served in less than 23 minutes is

A. 25
B. 40
C. 50
D. 60

END OF PAPER

Please use this page for rough work